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Iron Precipitation and Secondary Plugging During Acid Removal of Hematite-Based Filter Cakes

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Abstract

Effective removal of hematite-based filter cakes is crucial for restoring formation permeability in open-hole completions. While hydrochloric acid (HCl) is commonly used for filter cake dissolution in carbonate reservoirs, it can also lead to secondary damage through iron precipitation. This study investigates both the dissolution efficiency of hematite in 10 wt.% HCl and the potential for secondary formation damage during acid-rock interaction. A multi-stage experimental workflow was employed, including solubility testing, inductively coupled plasma (ICP) analysis during calcite addition, and core flooding simulations using Indiana limestone. Solubility results showed that HCl dissolved 45% of the hematite under dynamic conditions at 250°F. ICP analysis revealed significant iron precipitation as the acid became neutralized during reaction with calcite, particularly when pH exceeded 1.5. CT scan and permeability measurements confirmed that while acid soaking improved near-wellbore conditions with a 96% recovery in permeability after the primary damage, the far-zone experienced a 71% reduction in permeability compared to the original undamaged value due to secondary plugging. These results highlight the dual role of HCl in both mitigating and potentially redistributing damage, emphasizing the importance of managing acid volume, pH, and iron stability during filter cake removal treatments.

Introduction

Drilling fluids play an indispensable role in the success of oil and gas well construction. These engineered fluids facilitate several critical functions, including maintaining hydrostatic pressure, transporting drill cuttings to the surface, cooling and lubricating the bit, and stabilizing the wellbore (Shokry, Basfar, and Elkatatny 2024; Basfar et al. 2025). However, during overbalanced drilling, the differential pressure between the wellbore and the formation drives fluid and solids invasion into the reservoir (Bageri et al. 2021; Ridha et al. 2020). To mitigate this, a filter cake forms on the wellbore wall as solid particles in the mud deposit across permeable zones (Youssef et al. 2025; Shad et al. 2021). This filter cake acts as a barrier that restricts

further fluid invasion, thereby protecting the near-wellbore region from extensive damage (Addagalla et al. 2016; Abraham et al. 2025).

While essential during drilling, the filter cake becomes an impediment to production if not properly removed before the well is placed onstream (Morales 2025; Al-Mutairi and Al-Dhufairi 2010). This is particularly critical in open-hole completions, where the wellbore is not lined with casing, and reservoir fluids must flow directly through the exposed formation into the well (Fink 2021; Johnson, Ashton, and Nguyen 1992). Any residual filter cake in such scenarios can severely restrict inflow or injection, resulting in increased skin factor and reduced productivity (A. M. Al Moajil and Nasr-El-Din 2014; A. Al Moajil et al. 2008). Effective filter cake removal is, therefore, a necessary step in restoring the natural permeability of the formation and ensuring efficient well performance (Hu, Hu, and Weng 2025; Chellappah, Tarleton, and Wakeman 2010).

Various strategies have been developed for filter cake removal, including mechanical scraping, jetting, and chemical dissolution (Fink 2021; Li et al. 2019; Ruslim et al. 2007). Among chemical methods, hydrochloric acid (HCl) is commonly employed—especially in carbonate reservoirs—due to its ability to dissolve both filter cake materials and acid-soluble formation components (Siddig Ali, Alafnan, and Elkatatny 2022). The reaction between HCl and carbonate minerals enhances porosity and connectivity, offering the dual benefit of cleanup and stimulation (Al-Mutairi and Al-Dhufairi 2010; Wilson et al. 2022). However, when the filter cake contains acid-soluble weighting materials, such as hematite (Fe_2O_3), acid treatment can introduce new challenges (Assem, Nasr-El-Din, and De Wolf 2013).

Hematite, an iron oxide, is increasingly used as a weighting material in water-based drilling fluids (Tehrani et al. 2014). Its high specific gravity ($\sim 5.0 \text{ g/cm}^3$) allows for higher mud weights with lower solid volume compared to traditional materials like barite (Falode 2020). Additionally, hematite's acid solubility makes it theoretically removable by acids such as HCl (Siddig Ali, Mahmoud, and Elkatatny 2020). As HCl reacts with hematite, it produces ferric chloride (FeCl_3), significantly increasing the concentration of dissolved iron in the solution (Siddig Ali, Alafnan, and Elkatatny 2022). Upon acid spending or pH increase, ferric ions may hydrolyze and precipitate as iron oxyhydroxides (Assem, Nasr-El-Din, and De Wolf 2013; Almubarak et al. 2021; Hall and Dill 1988). These precipitates can plug pore throats and reduce permeability, counteracting the intended benefit of filter cake removal (Addagalla et al. 2016; Almubarak et al. 2021).

Secondary damage caused by iron precipitation during acidizing is a well-known issue in carbonate acid treatments (Assem, Nasr-El-Din, and De Wolf 2013). The risk is heightened in hematite-weighted mud systems, where the quantity of iron introduced into solution is much higher than in conventional barite-weighted systems (Siddig Ali, Alafnan, and Elkatatny 2022).

The present study investigates the formation damage induced by HCl-based removal of hematite-weighted filter cake under realistic reservoir conditions. The approach combines two complementary experimental techniques. First, a solubility and ICP analysis was conducted to evaluate the dissolution efficiency of hematite in HCl and the precipitation tendency of iron as acid interacts with carbonate material. Second, a core flooding experiment was designed to study the fluid–rock interaction in a carbonate core. The objective is to provide a more accurate depiction of both primary and secondary formation damage during a typical open-hole cleanup treatment using HCl.

Methodology

This study employed a multi-stage experimental approach to evaluate the formation damage associated with the removal of hematite-based filter cake using hydrochloric acid (HCl). The workflow included: (1) solubility testing of hematite in HCl, (2) investigation of acid-rock interaction dynamics via ICP analysis, and (3) core flooding tests to simulate primary and secondary damage mechanisms under realistic field conditions. Each phase was designed to characterize the efficiency of HCl and its potential to induce secondary damage due to iron precipitation or particle migration.

Solubility Test

To evaluate the solubility of hematite in hydrochloric acid prior to acid-rock interaction, a dissolution test was performed under dynamic conditions. A 10 wt.% HCl solution was prepared by diluting concentrated hydrochloric acid with deionized water, and 150 mL of the solution was heated to 250°F using a hot plate to simulate reservoir temperature. Then, 15 grams of hematite powder were added, and the mixture was continuously stirred at 400 RPM using a magnetic stirrer to ensure uniform acid-solid interaction. After 16 hours of reaction time, the mixture was filtered to separate the undissolved solids. The residue was subsequently dried and weighed to determine the dissolution efficiency, providing insight into the maximum amount of hematite that could be solubilized under these conditions.

Inductively Coupled Plasma (ICP) Analysis During Calcite Interaction

To simulate the interaction between spent acid and carbonate rock, the filtered solution saturated with dissolved iron was reacted with calcite powder. The tests were conducted at 250°F to replicate downhole temperature conditions. Calcite was incrementally added in four equal phases, with 6 grams introduced per 150 mL of iron-rich solution at each stage (equivalent to 4 g per 100 mL). After each addition, samples were collected at three time intervals: 10 minutes, 30 minutes, and 2 hours. The concentrations of Fe and Ca in the solution were analyzed using ICP spectroscopy, while the pH evolution was continuously monitored. This phase was designed to assess iron instability, calcium release, and the potential for precipitation during progressive neutralization by carbonate minerals.

Drilling Fluid Formulation and Properties

A water-based drilling fluid was formulated to build up the filter cake on the core face. The mud system incorporated hematite as the primary weighting material at a concentration of 350 g/L, aiming to achieve a high-density fluid that replicates realistic wellbore conditions. To enhance the fluid's performance, several additives were included in the formulation. Xanthan gum and bentonite were used to control viscosity, while PAC-R and starch were added to reduce fluid loss. Soda ash and potassium hydroxide (KOH) were employed to adjust the pH and manage calcium ion concentration. Additionally, a defoamer was introduced to minimize surface tension. The final mud had a density of 15.2 ppg, a plastic viscosity of 32 cP, and a yield point of 25 lb/100 ft². The filtrate volume under standard API test conditions was 10 mL/30 min, and the filter cake thickness was ~2.5 mm. These properties were chosen to ensure effective filter cake deposition and simulate realistic drilling-induced damage in carbonate formations.

Core Flooding Experiment

A core flooding setup was used to simulate the full sequence of filter cake deposition, removal, and acid-rock interaction. Indiana limestone cores were selected due to their homogeneous calcite content and well-characterized petrophysical behavior. Each core was first characterized for porosity and permeability using brine.

The hematite-weighted mud was then statically exposed to the core under overbalanced pressure, allowing filter cake deposition and internal solids invasion—representing primary formation damage. After cake formation, the core was removed, and the external cake was carefully scraped and dissolved in 10 wt.% HCl. This pre-dissolved solution was then injected into the core at half-pore volume, followed by a 12-hour soak under static conditions to represent the recommended acid contact time.

Subsequently, the squeeze scenario was simulated by flushing the core from the same direction as the initial injection, allowing the acid to penetrate through the second half of the core. This approach aimed to study both the cleanup efficiency and the potential for secondary damage due to iron precipitation or particle displacement.

Core Characterization Techniques

The core was analyzed before and after each phase using a combination of imaging and flow measurement methods. Computed Tomography (CT) scanning was employed to provide internal structural imaging of the core, allowing for the detection of features such as wormholes, densification zones, and the movement of solids. In addition, permeability testing was performed using steady-state brine flow to quantify any flow impairment or recovery throughout the experimental process.

After completing the flooding and acid treatment stages, each core was carefully cut along its length into two equal segments to differentiate between the near-wellbore region (initially exposed to drilling fluid and acid) and the far-zone (subjected primarily to the post-soak squeeze phase). Each half was independently characterized using CT scanning and liquid permeability measurements. This segmentation approach enabled spatial mapping of damage distribution and helped distinguish between primary damage effects near the core face and secondary damage trends resulting from fluid migration and precipitation deeper into the formation.

Results and Discussion

This section presents the results of formation damage evaluation during the removal of hematite-based filter cake using 10 wt.% hydrochloric acid (HCl). The investigation was carried out through a sequential experimental workflow involving three key stages: hematite solubility assessment, reaction monitoring with carbonate rock using ICP analysis, and a core flooding experiment simulating filter cake buildup, acid soaking, and squeeze treatment. This integrated approach allowed for simultaneous observation of both primary formation damage due to drilling fluid invasion and secondary damage resulting from acid-rock interaction and iron precipitation. The findings are interpreted using solubility metrics, ion concentration trends, CT scan profiles, and segmented permeability measurements.

Hematite Solubility in Hydrochloric Acid

The results showed that 6.75 grams of hematite were successfully dissolved, corresponding to a dissolution efficiency of 45% by weight. This level of solubility suggests that a single-stage HCl treatment could be sufficient to remove hematite-based filter cake in open-hole completions. For example, in a wellbore with an 8.5-inch diameter and a 2/32-inch-thick cake composed entirely of hematite, the tested removal solution has enough capacity to dissolve the entire cake volume without the need for repeated acidizing stages. In contrast, other solutions with more limited solubility may require multiple treatments to achieve comparable removal efficiency.

The dissolution behavior is closely linked to the physical characteristics of the hematite particles. Particle size analysis of the weighting material revealed a broad distribution, with a dominant fine fraction that enhances total surface area and facilitates rapid acid access. This mix of fine and coarse particles supports both effective cake formation during drilling and favorable reactivity during acid treatment. A plot of the particle size distribution is shown in [Figure 1](#), illustrating the suitability of the selected hematite for efficient dissolution.

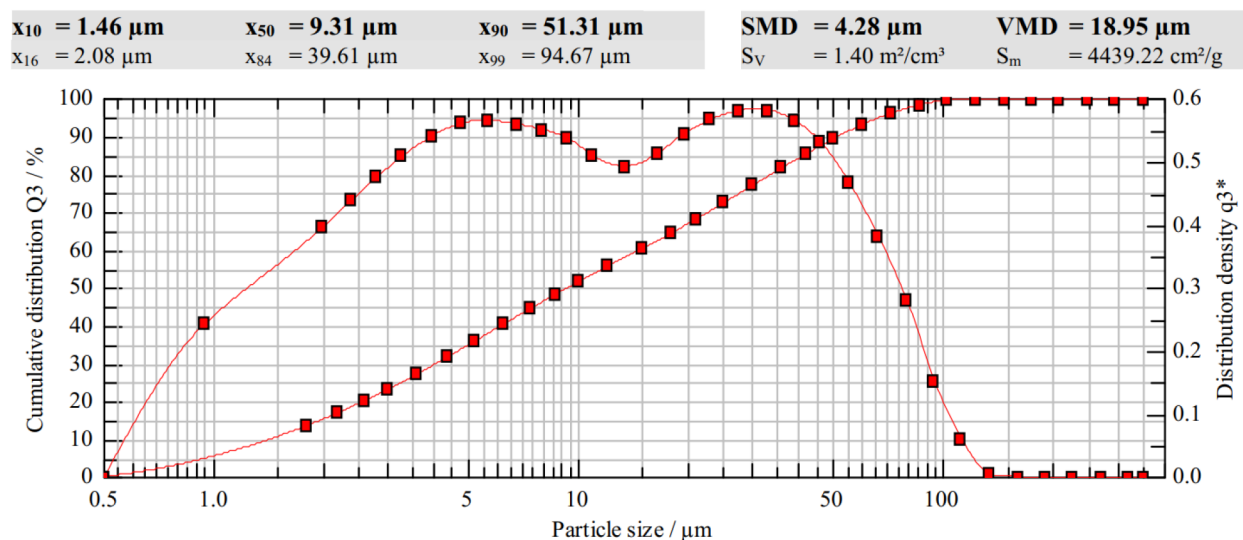


Figure 1—Particle size distribution of the hematite weighting material used in the solubility test

Reaction Dynamics During Calcite Addition

Following the solubility test, the filtered solution—containing dissolved iron from the hematite in 10 wt.% HCl—was subjected to stepwise interaction with powdered calcite to simulate acid-rock contact in carbonate reservoirs. The goal was to evaluate how the acid's reaction with carbonate impacts the stability of dissolved iron and whether secondary precipitation occurs under progressively neutralized conditions.

The results showed that calcium concentration consistently increased with each phase of calcite addition, indicating ongoing dissolution of the carbonate and active consumption of acid. In contrast, iron concentration remained stable during the first two phases but began to drop significantly during the third phase—coinciding with a measurable increase in pH beyond 1.5 as observed in Figure 2-a. By the end of the fourth phase, iron concentration had dropped close to zero, suggesting extensive iron precipitation as the acid became increasingly neutralized.

This behavior reflects the common instability of ferric ions (Fe^{3+}) in solution once pH exceeds critical thresholds (Taylor, Nasr-El-Din, and Al-Alawi 1999). In highly acidic conditions, Fe^{3+} remains soluble as FeCl_3 complexes; however, as H^+ ions are consumed through calcite reaction, hydrolysis of ferric ions leads to the formation of insoluble iron oxyhydroxides, which readily precipitate and pose a risk of secondary formation damage.

Visual inspection of the samples collected confirmed these trends as shown in Figure 2-b. In the early phases, the fluid remained clear with only minor discoloration, while later phases showed increasing turbidity and visible solid formation—particularly after the third calcite increment. This progression confirms that precipitation was not instantaneous but was strongly influenced by the acid's depletion and corresponding pH shift.

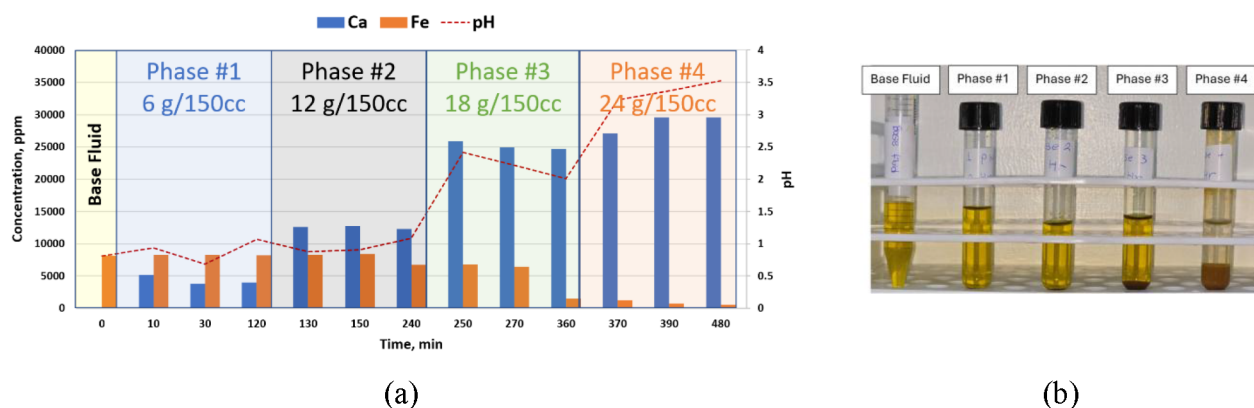


Figure 2—Variation in calcium and iron concentrations, along with pH changes, during stepwise addition of calcite to the HCl-hematite solution (a), Fluid samples collected at the end of each phase (b)

CT Scan-Based Evaluation of Damage Evolution

To evaluate the spatial distribution of formation damage and the effectiveness of filter cake removal, CT scan imaging was conducted at three key stages of the core flooding experiment: initial core condition, after filter cake buildup, and after acid treatment. The CT profiles provide a cross-sectional view of density changes throughout the core, helping to distinguish between primary and secondary damage mechanisms.

The initial scan, represented by the blue curve as shown in Figure 3, was performed before any fluid exposure and serves as the reference for undamaged carbonate rock. Following this, the core was exposed to the hematite-weighted drilling fluid described earlier, and a filter cake was successfully formed across the rock surface. The core was then extracted and scanned again, with the resulting CT profile plotted as the red curve. This scan clearly shows a marked increase in CT number along the first 22 mm of the core (corresponding to the first 22 slices), indicating the extent of solids invasion and formation damage caused by filter cake buildup. Beyond this region, the red and blue curves nearly overlapped, confirming that the damage was primarily concentrated near the inlet face.

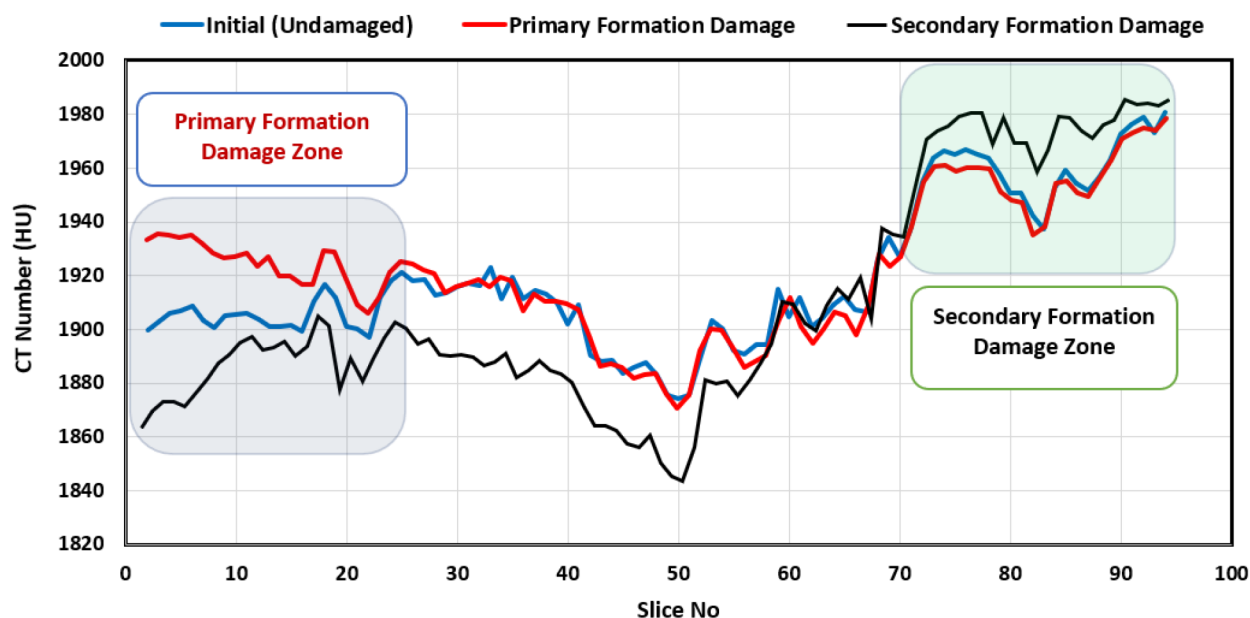


Figure 3—CT profiles showing initial (blue), primary damage (red), and post-acid treatment (black)

The final stage involved exposing the damaged core to a half-pore volume of HCl solution saturated with dissolved hematite, simulating a realistic post-cleanup environment. The acid was allowed to soak statically

for 12 hours, followed by a squeeze phase where the fluid was flushed in the same direction. A third CT scan was performed, and the profile is shown as the black curve.

This final scan revealed a notable decrease in CT number at the core's inlet, reflecting dissolution of solids and partial recovery from the primary damage. The progressive reduction in CT number across the first half of the core indicates continued mineral dissolution and porosity enhancement during acid-rock interaction. However, toward the outlet end of the core, the CT number began to increase again, suggesting precipitation of solids—likely iron-based compounds—as the acid became neutralized. This observation is consistent with the final phases of the ICP analysis, where iron precipitation was detected upon rising pH and depletion of free acid.

These CT scan results confirm that while HCl can effectively mitigate initial formation damage near the injection point, improper volume design or lack of precipitation control can lead to secondary damage deeper into the formation, especially during the squeeze phase.

Permeability Evaluation

To quantify the impact of drilling fluid invasion and subsequent acid treatment on flow capacity, brine permeability measurements were conducted at three stages of the core flooding experiment: (1) prior to filter cake deposition, (2) after exposure to hematite-weighted drilling fluid (representing primary damage), and (3) after acid soaking and squeeze flushing (to evaluate recovery or secondary damage effects).

The initial permeability of the Indiana limestone core was measured using a low-salinity brine under steady-state conditions and recorded at 206.3 mD. Following mud exposure and filter cake formation, a significant reduction in permeability was observed, with values dropping to 52.2 mD, confirming the effectiveness of the mud system in generating near-wellbore damage. This permeability drop corresponded closely with the CT scan results, which showed increased density and solid deposition across the first 22 mm of the core face.

After acid treatment—consisting of soaking the core with half a pore volume of 10 wt.% HCl solution saturated with dissolved hematite for 12 hours, followed by a squeeze flush—permeability was reassessed. The results showed partial recovery, with an overall post-treatment permeability of 75.3 mD. When the core was segmented into two equal halves for localized assessment, the first half (near-wellbore), exposed during the soak stage, recovered significantly to 105 mD, indicating effective dissolution of the filter cake and mobilized solids. Conversely, the second half (far-zone), contacted during the squeeze stage, showed a more modest permeability of 60 mD, suggesting secondary plugging possibly due to iron precipitation as the acid spent.

These trends, summarized in [Figure 4](#), highlight that while HCl treatment can restore permeability near the wellbore, improper fluid management may shift the damage deeper into the reservoir. This further reinforces the importance of controlling acid placement and minimizing iron instability in deeper zones during cleanup.

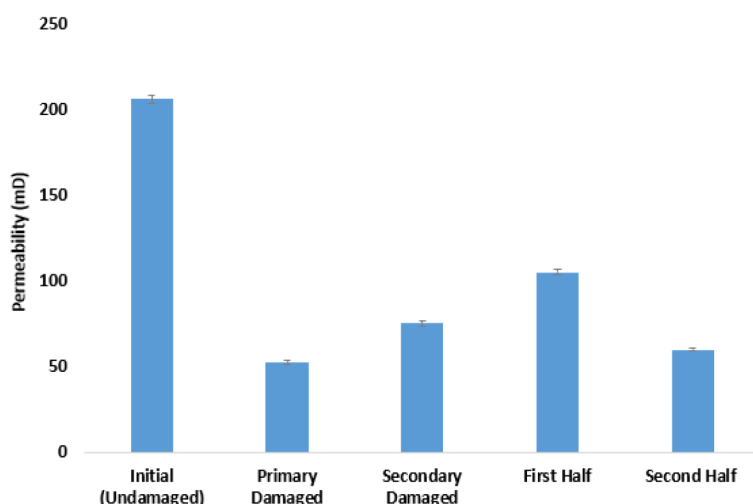


Figure 4—Permeability measurements at different stages of the experiment: initial (undamaged), after primary damage, after acid treatment, and for the segmented near-wellbore and far-zone sections

Conclusions

This study evaluated the effectiveness of hydrochloric acid (HCl) in removing hematite-based filter cakes from carbonate formations and investigated the formation damage that may occur during the treatment. A stepwise experimental approach was used, including solubility testing, reaction monitoring with calcite, and core flooding under realistic reservoir conditions. The goal was to understand both the benefits of acid treatment in restoring permeability and the risks of secondary damage due to iron precipitation deeper in the formation.

The key findings of this research can be summarized as follows

- Hydrochloric acid (10 wt.%) was able to dissolve about 4.5 g of hematite under realistic temperature conditions (250°F), making it a promising solution for cleaning thin hematite-based filter cakes.
- When HCl reacted with carbonate rock, the solution's pH increased, causing iron to precipitate. This can create new blockages in the formation and reduce permeability.
- Core flooding experiments showed that permeability near the wellbore recovered from the primary damage effect, but deeper in the formation, permeability dropped due to iron precipitation during the acid squeeze stage.
- While HCl can remove hematite filter cake and reduce near-wellbore damage, improper design or uncontrolled acid-rock reactions can lead to serious secondary damage.

Declaration

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